

IoT Intrusion Detection: A Review of ML and DL-Based Approaches

Imane Rakine

Faculty of Science Ben M'sik
Hassan II university - EIGSICA
Casablanca, Morocco
imane0704rakine@gmail.com

Kamal El Guemmat

ENSET
Hassan II university
Casablanca, Morocco
k.elguemmat@gmail.com

Sara Ouahabi

Faculty of Science Ben M'sik
Hassan II university
Casablanca, Morocco
sara.ouahabi@gmail.com

Issam Atouf

Faculty of Science Ben M'sik
Hassan II university
Casablanca, Morocco
issamatouf@gmail.com

Mohamed Talea

Faculty of Science Ben M'sik
Hassan II university
Casablanca, Morocco
taleamohamed@yahoo.fr

Abstract—The world is becoming more and more tied to the Internet of Things (IoT) technology. It is present in almost every era: healthcare, agriculture, education, industry, etc. This technology has many advantages, but also many challenges. The characteristics of IoT networks, such as heterogeneity, mobility, multi-tenancy, make the security of these networks a critical challenge. Many solutions have been proposed to ensure the security of IoT networks. One of the very interesting solutions is the intrusion detection system (IDS). Its main role is to continuously monitor the network traffic and detect any malicious activity. There are two intrusion detection strategies: signature-based detection (SIDS) and anomaly-based detection (AIDS). SIDS has demonstrated its effectiveness at detecting known attacks with low false alarms. However, it fails at detecting unknown and new attacks known as zero-day attacks. In contrast, AIDS has demonstrated effective results in detecting zero-day attacks. This article focuses on reviewing recent researches that present intrusion detection systems based on machine learning (ML) and deep learning (DL) techniques in IoT networks. The challenges faced by the reviewed articles are identified, and recommendations are suggested to give a clear path for future work.

Keywords—Internet of Things, Intrusion Detection System, Anomaly-based IDS, Machine Learning, Deep Learning, Zero-day attacks.

I. INTRODUCTION

The Internet of Things (IoT) technology is getting more and more widely present in various sectors related to our daily activities. IoT application sectors include healthcare, transport, agriculture, education, etc. In the healthcare sector, for example, biosensors are developed to be attached to human bodies for health records and remote patient monitoring [1]. In the education sector, campuses and classrooms are also becoming connected. As an example, student class attendance may be tracked using cards embedded with Radio-Frequency Identification (RFID) tags instead of the traditional way [2]. Homes, cities, vehicles, industries, and many other environments are becoming connected. It facilitates human life as it saves cost and time, automates maintenance, and speeds up processes.

The basic phases of the IoT network architecture are collection, transmission, and processing [3]. The collection phase consists of collecting the data from the sensor nodes and

aggregating it to be ready for transmission in the second phase to the specific destination (users and applications) [3]. Then, the necessary details are extracted, and actions are taken in the data processing phase to enable applications to communicate with the devices running on the network [3].

As the number of connected devices grows, a variety of IoT challenges appear and grow day by day. Moreover, the characteristics of IoT networks, including scalability, multi-tenancy, heterogeneity, mobility and device sizes, make them distinct from traditional networks and pose specific security requirements. One of the critical challenges for IoT researchers is ensuring security. Unknown and new attacks, known as zero-day attacks, are appearing every day, and threatening the security of each layer in the IoT system. Intrusion detection system (IDS) solutions, specifically anomaly-based intrusion detection systems (AIDS), have proven their effectiveness for detecting already known attacks, as well as zero-day attacks. The present article reviews research that focuses on machine learning (ML) and deep learning (DL)-based IDS for IoT network security.

The remaining content of this paper is divided into the following sections: Section II introduces the detailed IoT architecture, consisting of five layers. Then, intrusion detection system types and strategies are presented in Section III. Section IV provides a presentation of various recent works that address intrusion detection systems based on ML and DL techniques suggested for IoT networks. Then, the reviewed papers are discussed in Section V, and recommendations are suggested. In Section VI, we present a brief set of conclusions based on the identified challenges.

II. INTERNET OF THINGS

IoT technology refers to a set of various connected devices capable of communicating and sharing data [4]. There is no standard IoT architecture for all purposes, given that IoT devices are diverse and heterogeneous. Three layers form the basic IoT network architecture: the perception layer, the network layer, and the application layer. These three layers represent the core concept of IoT network technology [5]. However, it does not define the finer aspects of the IoT concept [6].

For finer aspects, a detailed IoT architecture consisting of five layers was proposed: perception, transport, processing, application, and business layer.

A. Perception layer

Also referred to as the object layer or sensor layer [7], [8]. It consists of sensors and actuators [9]. The main task done in this layer is collecting the environment's data using sensors. Then, the actuators take actions based on the collected data [9].

This layer requires more security attention, given that it has more interaction with the end user. It is vulnerable to many physical attacks, including node capture attacks, where the attacker succeeds in managing the sensor nodes [7]. Node tampering attacks aim to find the secret keys used for data decryption by hacking the system [7]. A code injection attack refers to injecting malicious code into sensor nodes to change the course of execution [7]. Disturbing RFID signals where the attacker tries to obstruct the transmission and reception of data by creating noisy signals in the network to make the services unavailable [10].

B. Network layer

Also referred to as the backbone layer. Here, the data collected in the perception layer is communicated and transferred to the next layer (processing layer) [8]. The transmission is done using communication technologies such as WIFI, Bluetooth, 3G, etc. The choice of communication technology depends on the device's constraints and features [10].

This layer is vulnerable to attacks that threaten the availability of network resources, such as a routing attack that causes a deviation of the traffic routing to an inappropriate path [9]. A man-in-the-middle attack occurs when an unauthorized user succeeds in interfering with the communication of two parties. It may manifest in different ways, including email or session hijacking, WIFI eavesdropping, etc [7]. Denial of service (DoS) and distributed denial of service (DDoS) attacks aim to saturate the services by sending spoofed requests until they become unavailable. Unlike DoS attacks, DDoS attacks are launched by multiple malicious nodes [7].

C. Processing layer

Also called the service management, or Middleware layer [8]. Its main role is to associate hardware devices with applications [11].

D. Application layer

Application layers visualize and personalize high-quality smart services for end users, such as temperature measurements, air humidity measurements, etc [8].

This layer is vulnerable to various attacks, including phishing attacks that aim to steal users' confidential information, often by sending malicious programs as email attachments [3]. It is also vulnerable to malware as well, which refers to any kind of malicious software, and it has many forms, such as ransomware, viruses, worms, Trojan horses, spyware, etc.

E. Business layer

In this layer, the entire system's activities and services are managed and regulated [8]. System management can be

related to users' authentication, data security, business models, applications, etc [12].

These security attacks and vulnerabilities can be avoided by utilizing defensive measures, including intrusion detection systems. The next section gives an overview of IDS types and strategies.

III. OVERVIEW OF IDS

IDS is a software or hardware system developed for monitoring a computer system or network traffic to detect intrusions [13]. An intrusion can be explained as an unauthorized activity that threatens the security of an information system [14].

The main function of an IDS is to assure the security of the information systems against any threat that may affect their confidentiality, integrity, and availability [15]. It monitors network traffic and detects malicious activities originating either from inside or outside the network [16].

Considering the source of the monitored data, IDS can be classified into two types: host-based IDS (HIDS), which is limited to inspecting data that originate only from the host system where it is installed; and network-based IDS (NIDS), responsible for monitoring the traffic that originates from all computers that belong to the same network where it is installed [14].

In general, there are two categories of intrusion detection systems based on the detection strategy adopted by the system.

A. Signature-based IDS (SIDS)

Also referred to as misuse identification. The signature-based strategy is efficient in detecting known attacks whose signatures match those of attacks registered in the SIDS database. However, it fails at detecting new and unknown intrusions [15].

B. Anomaly-based IDS (AIDS)

The Anomaly-based strategy succeeds at detecting unknown attacks [17]. AIDS's main techniques are statistics, knowledge, and machine learning approaches. The statistics-based detection approach requires building a model for a normal profile, which consists of a set of normal user behaviors, and then the system reports any deviation of the current traffic from the normal profile created [14]. The knowledge-based approach is based on defining a set of rules for normal system activity by an expert [14]. The system based on the machine learning approach is trained using either a labeled dataset known as supervised learning, or an unlabeled dataset, called unsupervised learning. Semi-supervised learning combines the use of labeled and unlabeled datasets for training the model. Several machine learning algorithms, such as Decision Tree (DT), Random Forest (RF), K-Nearest Neighbors (KNN), are used to classify attacks. However, in some complex cases, when dealing with high-dimensional datasets, deep-learning classifiers such as Auto-Encoder (AE), Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), are more efficient [18].

In the rest of this paper, we will focus on reviewing machine learning and deep learning-based intrusion detection systems suggested for IoT networks.

IV. REVIEWED PAPERS

This section presents recent research published between 2019 and 2024, which address ML/DL-based IDS as a solution for IoT network security. Table 1 summarizes the reviewed papers.

Hasan et al. [19] focused on ML techniques for developing their IDS model. They compared the performance of Logistic Regression (LR), Support Vector Machine (SVM), Artificial Neural Network (ANN), DT, and RF algorithms for multiclass classification, using collected data from Kaggle. The experimental results confirmed that the RF technique performed comparatively better, achieving an accuracy rate of 99.4%.

Roopak et al. [20] compared the performance of the following DL techniques for DDoS attack detection: CNN, LSTM, Multilayer Perceptron (MLP) and a combination of CNN and LSTM. They conducted the experiments on the CICIDS 2017 dataset. The CNN+LSTM combination technique outperformed the other classifiers, achieving an accuracy of 97.16%.

In [21], the authors tested single ML classifiers and ensemble ML classifiers for DoS attack detection. They used NSL-KDD, CICIDS-001, and UNSW-NB15 datasets for benchmarking Multi-Layer Perceptron (MLP) and Classification and Regression Trees (CART) single ML classifiers, as well as the following ensemble ML classifiers: Extremely Randomized Trees (ETC), AdaBoost (AB), Gradient Boosted Machine (GBM), RF, and Extreme Gradient Boosting (XGB). They concluded that XGB, CART, AB, MLP, and ETC can be used when high accuracy rate and low false positive rate (FPR) are the priorities of the IoT application, while CART, RF, and XGB are better when a quick response time is the application priority. In general, CART and XGB models demonstrated the optimal trade-off between detection performance and response time on all datasets.

A Random Forest-based intrusion detection system with a minimal feature set is proposed in [22]. In this study, only 10 features were manually selected among 41. The experiments reported accuracy rates of 98.1% and 98.82% using the NSL-KDD and KDD-CUP99 datasets, respectively. Moreover, a low FPR and reduced learning time were recorded.

A multi-stage optimized NIDS framework based on ML is suggested in [23]. The authors used the Synthetic Minority Oversampling technique (SMOTE) for balancing the dataset classes in the data preprocessing phase. Then, they investigated information gain (IG) and correlation-based (CB) techniques for selecting useful features. For the classification, they chose KNN and RF classifiers, and tested three distinct hyper-parameter tuning/optimization models on two different datasets: CICIDS 2017 and UNSW-NB 2015. The experiments demonstrated that the RF classifier with Bayesian Optimization using the Tree Parzen Estimator (BO-TPE-RF) outperforms the other tested optimization techniques.

In [24], two lightweight hybrid models of IDS named RCNN and XCNN are presented. The RCNN model uses the RF algorithm for selecting the useful features, and the XCNN model uses the eXtreme Gradient Boosting (XGBoost) algorithm. Both models are based on a CNN classifier. They evaluated the model in binary and multiclass classification scenarios using Balot and DoH20, publicly available datasets,

and their created dataset named CCD-INID-V1. The experiments demonstrated the effectiveness of the RCNN and XCNN techniques in detecting anomalies with reduced computation time compared to models based on the KNN classifier.

Rahman et al. [25] presented a centralized IDS approach based on the unsupervised auto-encoder (AE) technique for feature abstraction and the ANN algorithm for classification. Three wrapper-based feature methods (SVM, C4.5, and Naïve Bayes (NB)) were applied to select the meaningful features on the Aegean Wi-Fi Intrusion dataset. They compared the classification results obtained using the top 20 features selected by SVM, C4.5, and NB individually and those obtained using the features selected by the combination of the three methods. The authors reported that the feature combination method achieved the highest accuracy rate (99.95%). However, building this model requires the longest CPU operating time.

Keserwani et al. [26] presented a ML-based IDS named GWO-PSO-RF IDS. This model is developed using Grey Wolf Optimization (GWO) and Particle Swarm Optimization (PSO) techniques for selecting important and useful features, along with the RF classifier. Using KDD Cup 99, NSL-KDD, and CICIDS-2017 datasets, GWO-PSO-RF model achieved an accuracy of 99.88%, 99.24%, and 99.87%, respectively.

An IDS engine was presented in [27], based on the Deep Belief Networks (DBN) classifier. It was evaluated using TON_IOT_Weather dataset. An accuracy rate not exceeding 86.3% was recorded.

Khanday et al. [28] focused in particular on lightweight IoT networks when implementing IDS model for attack detection, especially DDoS attacks. They suggested an effective pre-processing technique. The initial step in the proposed pre-processing procedure is handling categorical data using the label encoding technique. Then, they performed data standardization using Standard Scaler, and for feature selection, they applied the ExtraTreeClassifier technique, which selected just 20 features from the BOT-IOT and TON-IoT datasets, used for evaluation. On each dataset, they tested the binary classification in two scenarios: one in which they considered all types of attacks available in the datasets with normal traffic, and the other in which they considered only DDoS attacks with normal traffic. The SMOTE technique was used for balancing the BOT-IoT dataset. They tested and presented the results of five classifiers: linear SVM, NB, LR, ANN, and CNN-LSTM combination.

Deep learning-based IDS models were evaluated in [29], where the SMOTE technique was used to balance the CIC IOT 2022 dataset classes and the Principal Component Analysis (PCA) technique for feature selection. The experiments were conducted in binary and multiclass classification scenarios using three classifiers: Feed Forward Neural Networks (FFNN), LSTM, and Random Neural Networks (RandNN). As a result, the FNN-based model outperformed the other models, attaining an accuracy rate of 99.93%.

Siganos et al. [30] developed an explainable intrusion detection system model. They proposed evaluating the following pre-trained ML and DL classifiers: NB, SVM, DT, RF, LR, XGBoost, AB, Quadratic Discriminant Analysis (QDA), and Deep Neural Network (DNN). Experiments were conducted on two different datasets: IEC 60 870-5-104 and CIC IoT dataset 2022. Recursive feature elimination and

sequential feature selection methods were performed for feature selection. They added the SHapley Additive exPlanations method for decision explanation to reinforce user confidence in the model. The RF classifier achieved the highest accuracy rate using the IEC 60 870-5-104 dataset, while the XGBoost classifier outperformed the other classifiers on the CIC IoT dataset 2022.

Khan et al. [31] proposed a hybrid DL-based IDS that can classify attacks across the three basic layers of the IoT architecture. The model was developed based on a combination of two DL classifiers: Recurrent Neural Network (RNN) and Gated Recurrent Units (GRU). They evaluated the effectiveness of this model using two optimization functions, Adam and Adamax. Through experiments on the ToN-IoT dataset for binary classification and multiclass classification, it was demonstrated that the proposed model outperforms previous IDS models, and indicated that Adam performs better than the Adamax optimization technique.

In [32], the authors demonstrated the effectiveness of regularized CNN models for detecting backdoor and scanning attacks, considered in this study as zero-day attacks. The experiments were done on the TON-IoT dataset using three ML classifiers (LR, NB, Adaboost), standard CNN, and regularized CNN classifiers. The experiments demonstrated that regularized CNN outperforms nonregularized CNN and

the other ML classifiers in detecting unknown attacks, achieving an accuracy rate of 87.59%.

Alaghbari et al. [33] studied the impact of implementing feature extraction techniques on the performance of different ML and DL algorithms in intrusion classification. Using N-BaIoT and MQTTset datasets, they compared the performance of RF, DT, XTree, NB, KNN, DNN, and CNN-LSTM classifiers, before and after applying the following methods for extracting the important and useful features: AE, PCA, linear discriminant analysis (LDA). The AE method demonstrated its effectiveness at extracting features. However, with some classifiers, it needs more training and testing time compared to LDA method.

Sharma et al. [34] developed an explainable DL-based IDS to overcome the challenge of explainability in DL techniques. The classifiers used in this study are DNN and CNN. They were evaluated using two different datasets. The correlation-based filter method was applied for feature selection. As a result, 36 features were selected from the NSL-KDD dataset and 38 from the UNSW-NB 15. These datasets are compatible with the DNN model. However, they are not compatible with the CNN model. They were transformed into 1D and 2D vectors for the 1D CNN and 2D CNN-based models, respectively. In general, DNN and 2D CNN achieved better results than the 1D CNN-based model on both datasets.

TABLE I. SUMMARIZATION OF RECENT PROPOSED ML AND DL-BASED IDS

References	Dataset	ML/DL classifier	Feature selection technique	Class balance technique	Output	Limitations
[19]	Simulated	LR, SVM, DT, RF, ANN	N/A	N/A	RF classifier outperformed the other classifiers with an accuracy of 99.4%.	The experiments were not based on real-time data
[20]	CICIDS2017	CNN, MLP, LSTM, CNN+LSTM	N/A	Duplicating method	CNN+LSTM combination technique outperformed the other DL classifiers, achieving an accuracy of 97.16%.	The time needed to train and test the system was not considered in the comparison.
[21]	CIDDS-001, UNSWNB15, NSL-KDD	RF, AB, GBM, XGB, ETC, CART, LMP	N/A	N/A	Some classifiers can be used when high accuracy and low FPR are the application's priority, while others are better when quick response time is the priority of the IoT application.	Only supervised ML algorithms are considered.
[22]	NSL-KDD, KDD CUP 99	RF	Manuel	N/A	The accuracy rate achieved is 99.9%, with reduced time and energy.	Unreal and outdated dataset.
[23]	CICIDS 2017, UNSW-NB 2015	KNN, RF	IG CB	SMOTE	-IGBFS method reached better detection accuracy and lower training time compared to the CBFS. -BO-TPE-RF technique outperformed the other classifiers.	Deep Learning techniques are not considered.
[24]	Created data, Balot, DoH20	CNN	RF XGBoost	N/A	The proposed model gives good results in less training time.	The created dataset should include more use-case scenarios to make it more realistic.
[25]	Aegean Wi-Fi Intrusion	ANN	SVM, C4.5, NB	N/A	The accuracy rate achieved exceeds 99.95%.	Requires long CPU time.

References	Dataset	ML/DL classifier	Feature selection technique	Class balance technique	Output	Limitations
[26]	KDDCup99, NSLKDD, CICIDS-2017	RF	GWO-PSO	Oversampling	The model reached on average 99.66% of accuracy rate using the three datasets.	-Outdated dataset used. -The time needed to train and test the model was not considered.
[27]	TON_IOT_Weather	DBN	N/A	N/A	An accuracy of 86.3% was achieved.	The experiments were not based on real-time data.
[28]	BOT-IOT and TON-IoT	SVM, NB, LR, ANN, CNN-LSTM	ExtraTreeClassifier	SMOTE	The results of the IDS experiments vary depending on the dataset, the classifier used for evaluation and the scenario of detection (Normal/All attacks or Normal/DDoS).	The time needed to train and test the model was not considered in the comparison.
[29]	CIC-IoT22	FFNN, LSTM, RandNN	PCA	SMOTE	FFNN outperformed the other classifiers.	The time required to train and test the model was not considered.
[30]	CIC-IoT-Dataset2022, IEC69870-5-104	SVM, DT, RF, XGBoost, AB, LR, QDA, DNN	-Recursive feature elimination -Sequential feature selection	N/A	-RF classifier outperformed the other classifiers on the IEC 60870-5-104 dataset. -XGBoost classifier outperformed the other classifiers on the CIC IoT dataset 2022.	The training and testing time was not considered when comparing the performance of the classifiers.
[31]	ToN-IoT	RNN-GRU	ReliefF	SMOTE	The proposed system outperformed previous models, achieving an accuracy of 99%.	-The dataset used does not capture all attacks that occur in real-world scenarios. -The required training and testing time was not considered.
[32]	TON-IoT	CNN	N/A	N/A	Regularized CNN performed better than the standard CNN and classical ML methods.	Limited amount of data for specific attacks.
[33]	N-BaIoT, MQTTset	DT, RF, XTree, KNN, NB, DNN, CNN-LSTM	AE, PCA, LDA	N/A	Integrating feature reduction techniques enhance the performance of the classifiers, and reduce the needed time for training and testing an IDS model.	More training and testing time is required when using AE feature reduction algorithm compared to LDA technique.
[34]	NSL-KDD, UNSW-NB15	CNN, DNN	Pearson correlation	N/A	The training time of the DNN-based system is shorter than that of the CNN model.	Data not balanced.

V. DISCUSSION AND FUTURE DIRECTIONS

One of the main issues with IoT networks is device limitation. IoT devices are often challenging in terms of computing resources. They often have limited memory space and energy-saving capacity. Moreover, IoT devices are often used in situations where it is not possible to recharge them frequently. Researchers in [24] and [28] have focused on overcoming these limitations that characterize lightweight IoT networks, and have suggested lightweight IDS process models. The feature selection technique is a key step in the detection process of a lightweight IDS. It aims to select the most meaningful and significant features and eliminate redundant and noisy data, which enhances the detection rate and reduces false alarms. Moreover, it reduces the time necessary for training and testing the system, and hence saving energy.

Another concern for security researchers is the complexity and ambiguity associated with AI-based IDS, particularly DL-based IDS. The process of DL models occurring between inputs and outputs lacks transparency and interpretability. The concept of explainable IDS is proposed in [30] and [34]. Integrating explainability tools into the system will help to build confidence and trust in the system, and improve its performance. They are responsible for communicating to the end-user an insight into how each feature contributes to the final result. By understanding the relationship between features and predictions, it is even possible to identify the weaknesses of these systems and significantly enhance their performance.

It is challenging to trust and rely on the results of experiments that have not been conducted in real-world scenarios to test the classification performance of an IDS. The authors in [19] used a simulated dataset to train and test their

model. However, real attack scenarios are more complicated and more challenging to deal with. Datasets such as NSL-KDD and KDDCup 99, used by the researchers in [21], [22], [34] and [26], do not cover IoT attack scenarios. They are outdated and do not include zero-day attacks. So, for efficient evaluation, it is recommended to use realistic datasets and conduct the experiments in real-time.

It is also recommended to conduct an in-depth study to define the IoT application's priorities (low FPR, real-time response, high accuracy, etc.) before choosing the appropriate classifier. The choice of the IDS classifier is of significant importance. As demonstrated in [21], some classifiers are better when quick response time is the IoT application's priority, and others are better when high accuracy and low FPR are the application's priority.

VI. CONCLUSION

IoT networks have many characteristics that distinguish them from traditional networks, such as scalability, heterogeneity, multitendency, mobility, limited memory space, limited energy-saving capacity, etc. Therefore, ensuring their security is a major concern for IoT researchers.

In this article, we started by providing an overview of IoT architecture, IDS types and detection strategies. Then we presented recent research published between 2019 and 2024 focusing on ML/DL-based intrusion detection systems as a solution for known and unknown attacks detection in IoT networks.

Based on the reviewed papers, we conclude that: various recent studies suggest lightweight IDS models as a solution for limited computing resource constraints in IoT networks. Other recent studies have also focused on the concept of explainable IDS, which enables users to understand the predictions and decisions of AI-based IDS models, especially those based on DL techniques. The simulated datasets and some available open-source datasets do not reflect realistic network traffic scenarios. Moreover, some datasets such as NSL-KDD are outdated and do not contain recent intrusions and attacks.

This review will enable security researchers to familiarize themselves with the key concepts of intrusion detection systems based on ML and DL solutions suggested for IoT networks.

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